# Accenture Material –C++ Question

## 1. What will be the output/effect of the following C++ code?

```
#include <iostream>
using namespace std;

int main() {
    int A[5][5], k, j;
    for (k = 0; k < 5; ++k)
        for (j = 0; j < 5; ++j)
            A[k][j] = A[j][k];
}
```

A. It transposes the given matrix A

B. It does not alter the given matrix

C. It makes the given matrix A symmetric

D. None of the above

Answer: A

## 2. What is the output of the following C++ code?

```
#include <iostream>
using namespace std;

int main() {
    long double a, b;
    int arr[sizeof(!a + b)];
    cout << sizeof(arr);
}
```

A. Run time error

B. 32

C. 64 with warning

D. No output

Answer: C

## 3. Which of the following statements is true regarding automatic/local variables in C++?

A. They give a reference to a global variable visible to all program files

B. They remain in existence for the lifetime of the program

C. Local variables have automatic storage duration by default

D. They are used to force local variables into a CPU register

Answer: C

## 4. What is the output of the following C++ code?

```
#include <iostream>
using namespace std;

int main() {
    int x = 4, y = 0;
    int z;
    z = (y++, y);
    cout << z << endl;
    return 0;
}
```

A. 1

B. 0

C. Undefined behavior due to evaluation order

D. Compilation error

Answer: A

## 5. What is the output for the following C++ code if the input is 1?

```
#include <iostream>
using namespace std;

int main() {
    int ch;
    cin >> ch;
    switch (ch, ch + 1) {
        case 1: cout << "1\n"; break;
        case 2: cout << "2\n"; break;
        default: cout << "3\n";
    }
}
```

A. 1

B. 3

C. 2

D. No output

Answer: C

6. What is the output for input 134?

```
int fun1(int num) {
    static int a = 0;
    if (num > 0) {
        a = a + 1;
        fun1(num / 10);
    } else {
        return a;
    }
}

int main() {
    cout << fun1(134);
}
```

A. 2

B. 3

C. Runtime error

D. None of these

Answer: B

7. What will be the output of the following C++ equivalent pseudocode for input 7?

```
int n;
cin >> n;
int m = 1, t = 0;
while (m < n) {
    t = t + m;
    m += 1;
}
cout << t;
```

A. 32

B. 76

C. 56

D. 28

Answer: D

8. What will be the output of the following C++ function for input 2?

```
int fun(int n) {
    if (n == 4)
        return n;
    else
```

```
        return 2 * fun(n + 1);
}
```

A. 4

B. 8

C. 16

D. Error

Answer: C

## 9. What will be the output?

```
int i = 5, j = 7;
if (i + j > 5) {
    j = i + 2;
    if (j < 5)
        cout << i;
    else
        cout << j;
} else {
    cout << i + 1;
}
```

A. 12

B. 5

C. 7

D. 6

Answer: C

## 10. What will be the output?

```
int j = 41, k = 37;
j = j + 1;
k = k - 1;
j = j / k;
k = k / j;
cout << k << " " << j;
```

A. 42 36

B. 36 1

C. 1 1

D. 1 36

Answer: D

## 11. What will be the output?

```
int a = 0, b = 1, c = 2;
*((a + 1 == 1) ? &b : &a) = a ? b : c;
cout << a << ", " << b << ", " << c;
```

A. 0 1 2

B. 0 2 0

C. 0 2 2

D. Error

Answer: C

## 12. Compare the outputs of the following two expressions:

```
int a = 40, b = 35, c = 20, d = 10;
cout << a * b / c - d << endl;
cout << a * b / (c - d);
```

A. Differ by 80

B. Same

C. Differ by 50

D. Differ by 160

Answer: A

## 13. What will be the output of the following two expressions?

```
int a = 60, b = 35, c = -30;
cout << (a > 45 || b > 50 && c > 10) << endl;
cout << ((a > 45 || b > 50) && c > 10);
```

A. 0 and 1

B. 0 and 0

C. 1 and 1

D. 1 and 0

Answer: D

## 14. What will be the output?

```
int a = 984, b = 10;
float c;
c = a / b;
cout << c;
```

A. 984

B. 98.4

C. 98

D. Error

Answer: C

## 15. Which condition will allow execution of Statement A?

```
if (condition1) {
    if (condition2) {
        // Statement A
    } else if (condition3) {
        // Statement B
    } else {
        // Statement C
    }
} else if (condition4) {
    // Statement D
} else {
    // Statement E
}
```

A. !condition2 && !condition3

B. condition1 && condition4 && !condition2 && !condition3

C. condition1 && condition2

D. !condition1 && condition2 && !condition4

Answer: C

## 16. What will be the output?

```
int num = 8;
cout << (num << 1) << " " << (num >> 1);
```

A. 8 0

B. 0 0

C. 16 4

D. Error: operation cannot be performed

Answer: C

## 17. What will be the output?

```
int i = 16;
i = !i > 15;
cout << "i = " << i;
```

A. i = -1

B. i = 0

C. i = 1

D. Error: undefined operation

Answer: B

## 18. What will be the output?

```
int x[10] = {0,1,2,3,4,5,6,7,8,9};
cout << sizeof(x);
```

A. 40

B. 10

C. 20

D. Error

Answer: A

## 19. What will be the output?

```
int x = 2;
(x & 1) ? cout << "true" : cout << "false";
```

A. true

B. false

C. 0

D. Error

Answer: B

## 20. What will be the output?

```
int a = 4, b = 2;
cout << "a^b = " << (a ^ b);
```

A. 4

B. 1

C. 0

D. 6

Answer: D

## 21. What will be the output?

```
int a = 4, b = 2;
cout << "a|b = " << (a | b);
```

A. 4

B. 1

C. 0

D. 6

Answer: D

## 22. What will be the output?

```
int a = 0 - true;
cout << a;
```

A. -1

B. Garbage value

C. 0

D. Error

Answer: A

## 23. What will be the output?

```
int x = 0;
if (x == x)
    cout << "if";
else
    cout << "else";
```

A. Two same variables cannot be compared

B. ifelse

C. else

D. if

Answer: D

## 24. What will be the output?

```
const int FALSE = -1;
const int NULL_VALUE = 0;
const int TRUE = 1;

if (NULL_VALUE)
    cout << "NULL";
```

```
else if (FALSE)
    cout << "TRUE";
else
    cout << "FALSE";
```

A. TRUE

B. FALSE

C. NULL

D. Error

Answer: A

## 25. What will be the output?

```
#include <iostream>
using namespace std;

int main() {
    if (true)
        cout << "work";
    else
        cout << "not work";
}
```

A. work

B. not work

C. Compiler error

D. Runtime error

Answer: A

## 26. What will be the output?

```
if (cout << "0")
    cout << "inside if block";
else
    cout << "inside else block";
```

A. inside else block

B. 0

C. 0inside if block

D. Error

Answer: C

## 27. What will be the output?

```
int i = 5, j = 4;
if (!(cout << ""))
    cout << i << " " << j;
else
    cout << i++ << " " << ++j;
```

A. 5 5

B. 5 4

C. 5 6

D. 6 6

Answer: B

## 28. What will be the output?

```
int i = 25;
if (i == 25);
i = 50;

if (i == 25)
    i = i + 1;
else
    i = i + 1;

cout << i;
```

A. 51

B. 25

C. 50

D. None of these

Answer: A

## 29. What will be the output?

```
if (sizeof(0))
    cout << "Hai";
else
    cout << "Bye";
```

A. 2

B. Bye

C. Runtime error

D. Hai

Answer: D

## 30. What will be the output in C++?

```
if (sizeof('\0'))
    cout << "inside if block";
else
    cout << "inside else block";
```

A. inside if block

B. inside else block

C. Null pointer exception

D. None of these

Answer: A

## 31. What will happen when this C++ code is compiled?

```
int i = 65;
switch (i) {
    case 65:
        cout << "Integer 65";
        break;
    case 'A':
        cout << "Char 65";
        break;
    default:
        cout << "Bye";
}
```

A. Integer 65

B. Char 65

C. Bye

D. Error: duplicate case values

Answer: D

## 32. What will be the output?

```
switch (2 / 3) {
    case 1:
        cout << "case 1 executed ";
    case 2:
        cout << "case 2 executed ";
        break;
    default:
        cout << "Default block executed";
}
```

A. case 1 executed

B. case 2 executed

C. Default block executed

D. Error: switch cannot hold this expression

Answer: C

## 33. What will happen when this C++ code is compiled?

```
int i = 1;
switch (i) {
    case i:
        cout << "case 1 executed";
        break;
    case i + 1:
        cout << "case 2 executed";
        break;
    default:
        cout << "default block executed";
}
```

A. case 1 executed

B. case 2 executed

C. default block executed

D. Compilation error: case labels must be constant expressions

Answer: D

## 34. What will be the output?

```
while (cout << 5 < 4)
    cout << "Loop ";
```

A. Loop Loop Loop Loop Loop

B. Infinite loop

C. 5Loop 5Loop 5Loop 5Loop 5Loop

D. None of these

Answer: B

## 35. What will be the output?

```
const int NULL_VALUE = 0;
while (NULL_VALUE == 0) {
    cout << "Loop";
```

```
    break;
}
```

A. Loop

B. Null

C. 0

D. Error: NULL cannot be compared

Answer: A

## 36. What will be the output?

```
float ft = 7.5;
while (ft) {
    cout << "Loop";
    ft = ft - 0.5;
    if (ft == 5.0f)
        break;
}
```

A. LoopLoopLoopLoopLoop

B. Loop

C. No output

D. None of these

Answer: A

## 37. What will be the output?

```
while (!!7)
    cout << "Hai";
```

A. Hai

B. HaiHai

C. Infinite loop

D. None of these

Answer: C

## 38. What will be the output?

```
while (!(cout << "awesome"))
    ;
```

A. awesome

B. Error

C. Infinite loop

D. None of these

Answer: A

## 39. In C++, if a line is read using std::getline(), what will the string contain?

```
std::string str;
std::getline(std::cin, str);
```

A. "I am a boy\\r\\n\\0"

B. "I am a boy\\r\\0"

C. "I am a boy\\n\\0"

D. "I am a boy"

Answer: D

## 40. What is the purpose of std::ios::binary in the following C++ statement?

```
std::ifstream fp("source.txt", std::ios::binary);
```

A. Open source.txt in binary mode for reading

B. Open source.txt in binary mode for reading and writing

C. Create source.txt for reading and writing

D. None of the above

Answer: A

## 41. What does the following C++ stream object represent?

```
std::ifstream fp("trial");
```

A. The first character in the file

B. An input file stream associated with the file

C. The name of the file only

D. The last character in the file

Answer: B

## 42. Which operations can be performed using the following C++ file stream?

```
std::fstream fp("NOTES.TXT", std::ios::in | std::ios::out);
```

A. Reading only

B. Writing only

C. Appending only

D. Reading and writing

Answer: D

## 43. How can the following C++ variables be printed?

```
float a = 3.14f;
double b = 3.14;
std::cout << a << " " << b;
```

A. Use cout with both variables

B. Use %Lf for both

C. Use printf with %Lf for both

D. Use %f only for double

Answer: A

## 44. Which statement correctly closes all three C++ file streams?

```
std::ifstream fs("A.C");
std::ifstream ft("B.C");
std::ofstream fp("C.C");
```

A. fp.close(fs, ft, fp);

B. fs.close(ft);

C. ft.close(fp);

D. fs.close(); ft.close(); fp.close();

Answer: D

## 45. What will be written to target.txt by this C++ program if source.txt contains "To err is human"?

```
std::ifstream fs("source.txt");
std::ofstream ft("target.txt");
char ch;
while (fs.get(ch)) {
    fs.seekg(4, std::ios::cur);
    ft.put(ch);
}
```

A. r n

B. Trh

C. err

D. None of the above

Answer: D

## 46. Which is the correct C++ way to read a float and a double?

```
float a;
double b;
```

A. cin >> a >> b;

B. scanf("%Lf %Lf", &a, &b);

C. scanf("%f %Lf", &a, &b);

D. printf("%f %lf", a, b);

Answer: A

## 47. Which is safer for reading a complete line in modern C++?

A. gets()

B. std::getline()

Answer: B

## 48. Which statement correctly obtains the underlying file descriptor of a C++ stream on a POSIX system?

```
std::ifstream fp("DUMMY.C");
```

A. It is obtained portably with fp.fileno()

B. A standard C++ ifstream has no portable fileno() member

C. fileno() returns the file size

D. cout automatically supplies the descriptor

Answer: B

## 49. What is the correct C++ approach when two output streams refer to the same file?

```
std::ofstream f1("file.txt");
std::ofstream f2("file.txt");
```

A. Both streams safely append independently

B. The second open may truncate/replace the file; behavior should not be used this way

C. The file automatically contains both writes in order

D. C++ forbids opening a file twice

Answer: B

## 50. What will be the output?

```
int k = 1;
std::cout << k << " == 1 is "
          << (k == 1 ? "TRUE" : "FALSE") << '\n';
```

A. k == 1 is TRUE

B. 1 == 1 is TRUE

C. 1 == 1 is FALSE

D. K == 1 is FALSE

Answer: B

## 51. What is the output of the following C++ code?

```
std::string str =
    "str = %c%s%c";
std::cout << "str = " << str;
```

A. str = %c%s%c

B. str = "str = %c%s%c"

C. No output

D. Error in program

Answer: A

## 52. If source.txt contains "Be my friend", what is printed?

```
std::ifstream fs("source.txt");
fs.seekg(0, std::ios::end);
fs.seekg(-3, std::ios::cur);

char c[5] = {};
fs.read(c, 3);
std::cout << c;
```

A. friend

B. frien

C. end

D. Error in fseek()

Answer: C

## 53. What will be the output?

```
double a = 3.15529;
std::cout << std::fixed << std::setprecision(1) << a;
```

A. 3.00

B. 3.15

C. 3.2

D. 3

Answer: C

## 54. What will be the output?

```
std::cout << char(~('C' * -1));
```

A. A

B. B

C. C

D. D

Answer: B

## 55. Suppose abc.txt contains "This is India". What will this C++ program print?

```
std::ifstream fp("abc.txt");
char ch;
while (fp.get(ch))
    std::cout << ch;
```

A. This is India

B. This is

C. Infinite loop

D. Error

Answer: A

## 56. What will be the output?

```
const char* p = "%d\n";
p++;
p++;
std::printf(p - 2, 23);
```

A. 21

B. 23

C. Error

D. No output

Answer: B

## 57. Which is the correct C++ approach to read the contents of a text file character by character?

```
std::ifstream ptr("myfile.txt");
char ch;
while (ptr.get(ch))
    std::cout << ch;
```

A. Print the contents of myfile.txt

B. Print only up to a NULL character

C. Infinite loop

D. Error in program

Answer: A

## 58. What will be the output?

```
std::cout << "%%" << '\n';
```

A. %%%%%

B. %%

C. No output

D. Error

Answer: B

## 59. What is the output of the following C++ expression?

```
int x[5][5]{};
for (int k = 0; k < 5; ++k)
    for (int j = 0; j < 5; ++j)
        x[k][j] = x[j][k];
```

A. It transposes the matrix

B. It does not alter the matrix

C. It makes the matrix symmetric

D. None of the above

Answer: A

## 60. What is the output?

```
long double a = 1.0L, b = 2.0L;
int arr[sizeof(!a + b)];
std::cout << sizeof(arr);
```

A. Runtime error

B. 32

C. 64 with warning

D. No output

Answer: C

## 61. Which statement best describes an ordinary local variable in C++?

A. It is always a global reference

B. It exists for the lifetime of the program

C. It normally has automatic storage duration

D. It is always stored in a CPU register

Answer: C

## 62. What is the output?

```
int x = 4, y = 0;
int z = (y++, y);
std::cout << z;
```

A. 1

B. 0

C. Undefined behavior

D. Compilation error

Answer: A

## 63. What is the result of this C++ switch expression for input ch = 1?

```
int ch = 1;
switch (ch, ch + 1) {
    case 1: std::cout << 1; break;
    case 2: std::cout << 2; break;
    default: std::cout << 3;
}
```

A. 1

B. 3

C. 2

D. No output

Answer: C

## 64. What is the output for input 134?

```
int fun1(int num) {
    static int a = 0;
    if (num > 0) {
        ++a;
        fun1(num / 10);
    } else {
        return a;
    }
    return a;
}
std::cout << fun1(134);
```

A. 2

B. 3

C. Runtime error

D. None of these

Answer: B

## 65. What is the time complexity of searching for an element in a circular linked list?

A. O(n)

B. O(n log n)

C. O(1)

D. None of the mentioned

Answer: A

## 66. In the worst case, how many comparisons are needed to search a singly linked list of length n?

A. $\log_2 n$

B. n/2

C. $\log_2 n - 1$

D. n

Answer: D

## 67. Which of the following has the best asymptotic performance?

A. O(n)

B. O(n!)

C. O(n log n)

D. O(n^c)

Answer: A

## 68. How many times is the following loop executed?

```
char ch = 'b';
while (ch >= 'a' && ch <= 'z')
    ++ch;
```

A. 0

B. 25

C. 26

D. 1

Answer: C

## 69. What is the auxiliary space required by this function if sizeof(int) = 2 bytes?

```
int sum(int A[], int n) {
    int sum = 0, i;
    for (i = 0; i < n; ++i)
        sum += A[i];
    return sum;
}
```

A. 2n + 8

B. 2n + 4

C. 2n + 2

D. 2n

Answer: A

## 70. What is the output for a = 8 and b = 9?

```
int function(int a, int b) {
    if (a < b)
        return function(b, a);
    else if (b != 0)
        return a + function(a, b - 1);
```

```
    else
        return 0;
}
```

A. 56

B. 88

C. 72

D. 65

Answer: B

## 71. What is the output?

```
int m = 9, n = 6;
m = m + 1;
n = n - 1;
m = m + n;
if (m > n)
    std::cout << m;
else
    std::cout << n;
```

A. 6

B. 5

C. 10

D. 15

Answer: D

## 72. What is the output?

```
int f = 6, g = 9, sum = 0;
for (int n = f; n < g; ++n)
    sum += n;
std::cout << sum;
```

A. 21

B. 15

C. 9

D. 6

Answer: A

73. A hash table has 9 slots and h(k) = k mod 9. Using chaining, keys 5, 28, 19, 15, 20, 33, 12, 17, 10 are inserted. What are the maximum, minimum and average chain lengths?

A. 3, 0, and 1

B. 3, 3, and 3

C. 4, 0, and 1

D. 3, 0, and 2

Answer: A

74. If quicksort always chooses the central element as pivot, what is the tightest upper bound for worst-case performance?

A. $O(n^2)$

B. O(n log n)

C. T(n log n)

D. $O(n^3)$

Answer: A

75. Let G have n vertices and m edges and be represented using an adjacency matrix. What is the tightest upper bound for DFS?

A. O(n)

B. O(m+n)

C. $O(n^2)$

D. O(mn)

Answer: C

76. Quicksort uses the first element as pivot. Let t1 and t2 be the comparisons for {1,2,3,4,5} and {4,1,5,3,2}. Which is true?

A. t1 = 5

B. t1 < t2

C. t1 > t2

D. t1 = t2

Answer: C

## 77. What does the following C++ function perform?

```
void func(Tree* root) {
    func(root->left);
    func(root->right);
    std::cout << root->data;
}
```

A. Preorder traversal

B. Inorder traversal

C. Postorder traversal

D. Level-order traversal

Answer: C

## 78. How can the minimum element be found in a binary search tree?

```
void min(Tree* root) {
    while (root->left != nullptr)
        root = root->left;
    std::cout << root->data;
}
```

A. Repeatedly move to root->left until it is nullptr

B. Repeatedly move to root until nullptr

C. Repeatedly move to root->right

D. Use the rightmost node

Answer: A